INCH-POUND

MIL-B-49430/7G(ER)
18 April 1994
SUPERSEDING
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MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5588/U

This specification sheet is approved for use within Army Research Laboratory and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

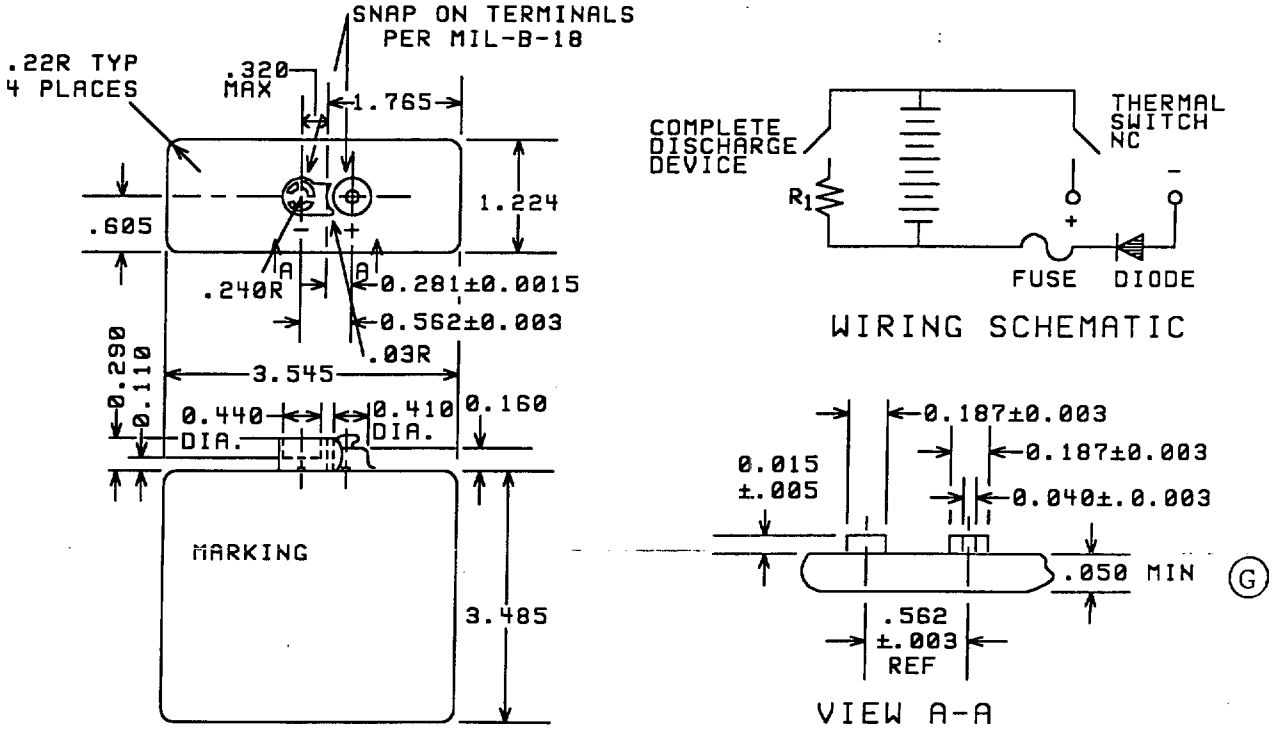


Figure 1. Battery dimensions and wiring schematic.

(G) denotes change

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NOTES:

1. All dimension are in inches.

2. Tolerances, unless otherwise specified, are $\pm$.015 inch.

3. The battery connector shall be snap-on terminals per MIL-B-18 with protective shield (negative terminal) as shown in Figure 1.

4. Terminal markings shall be printed or embossed on the jacket near the connector on the terminal face of the battery.

5. R_1 value shall be between 300 ohms and 430 ohms and have a power rating of 1.0 watt.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 15.25 volts.

Minimum voltage (for capacity): 11.00 volts.

The battery shall contain a 1 ampere non-replaceable slow blow fuse wired in series with cells in the negative leg.

The battery shall contain a diode wired in series to prevent charging. See paragraph 3.5.5.2 of the basic specification.

Terminals: Snap-on type (see Figure 1).

Weight (maximum): 0.65 pound (295 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 11.00 Volts
I	3.6
L	1.7
H	3.6
HT	3.1
LT	1.6
T	3.3
D	3.3

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Initial voltage delay: During the I, H, HT, T, and D discharge tests, initial voltages below the 11.00 volt minimum voltage cannot exceed a 10 second duration. During L discharge tests initial voltage below 10.00 volts cannot exceed a 12 second duration. During LT discharge tests initial voltage below 10.00 volts cannot exceed a 20 second duration.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery in the series circuit as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all specified capacity tests are described in the basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 0.8 ampere constant current at $70 \pm 5^{o}F$ for a minimum of 3.1 hours to 2.2 volt cutoff. Continue discharge to zero volts, then discharge at 0.4 ampere constant current for 3.1 hours into voltage reversal.

Batteries: For all capacity tests the batteries shall be discharged at 0.8 ampere constant current to zero volts. For the I capacity test the batteries shall be force discharged at 0.4 ampere constant current for 5.6 hours immediately after zero volts is obtained.

Closed-circuit voltage: Load resistance of 25 ohms shall be used between battery terminals. Voltage shall be above 11.00 volts within 10 seconds.

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